Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750–2024 Overview

Data: Global Carbon Budget 2025, NASA

Atmospheric CO_2 Reaches 429 ppm

50% Above Pre-Industrial Levels

429

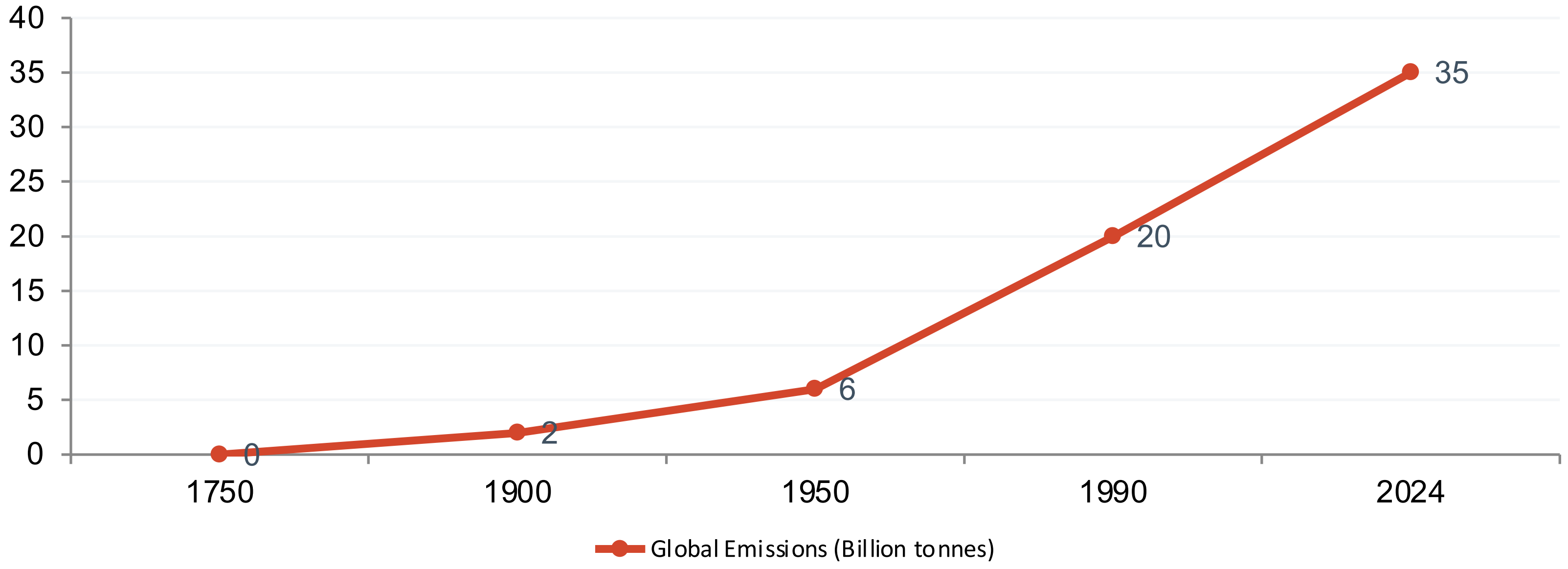
Parts per million (ppm)
Feb 2026 Measurement (NOAA)

Context

- • Pre-industrial baseline: ~280 ppm
- • At least 1.5x greater than natural increase after last ice age
- • Sharp acceleration observed since mid-20th century
- • Driven primarily by coal, oil, and natural gas

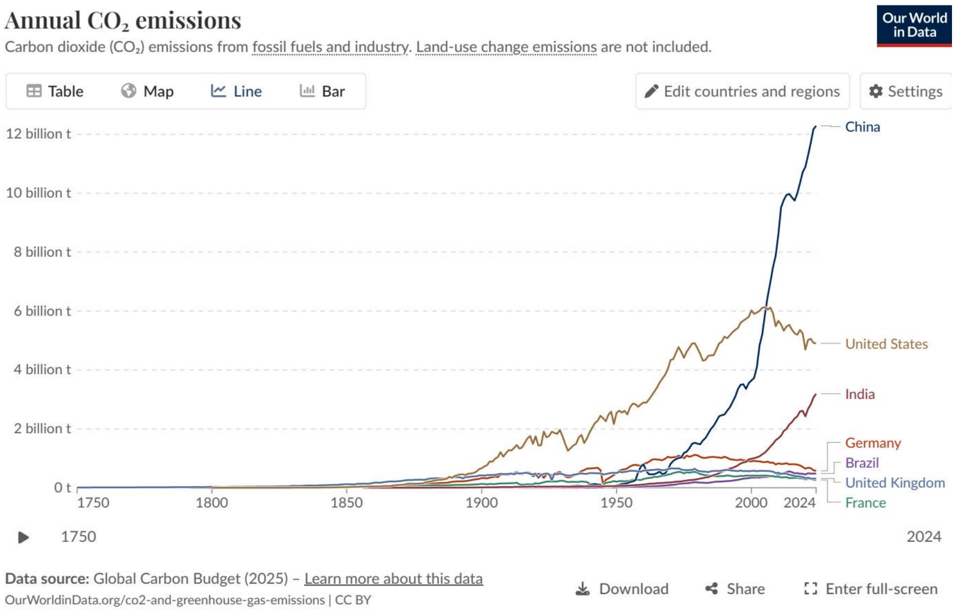
Global Fossil Fuel CO_2 Emissions Grew Six-Fold

From 6 Billion tonnes in 1950 to 35+ Billion Tonnes Today



China Now Emits Over 12 Billion Tonnes

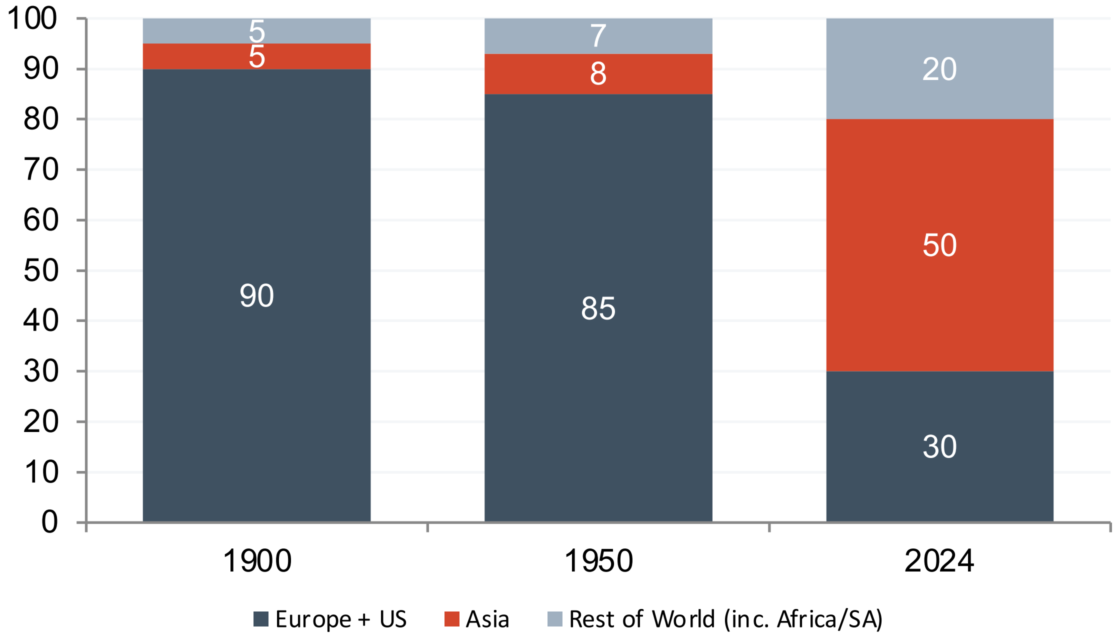
More Than Double the United States



Country / Region	Annual CO_2 (Bt)	Trend
China	~12.5 (~27%)	Slight increase
United States	~5.0 (~14%)	Declining
India	~3.0 (~8%)	Rising
EU-27	~2.7 (~7%)	Declining
Germany	~0.6 (~1.7%)	Declining
Brazil	~0.5 (~1.4%)	Stable

Asia Now Accounts for ~50% of Global Emissions

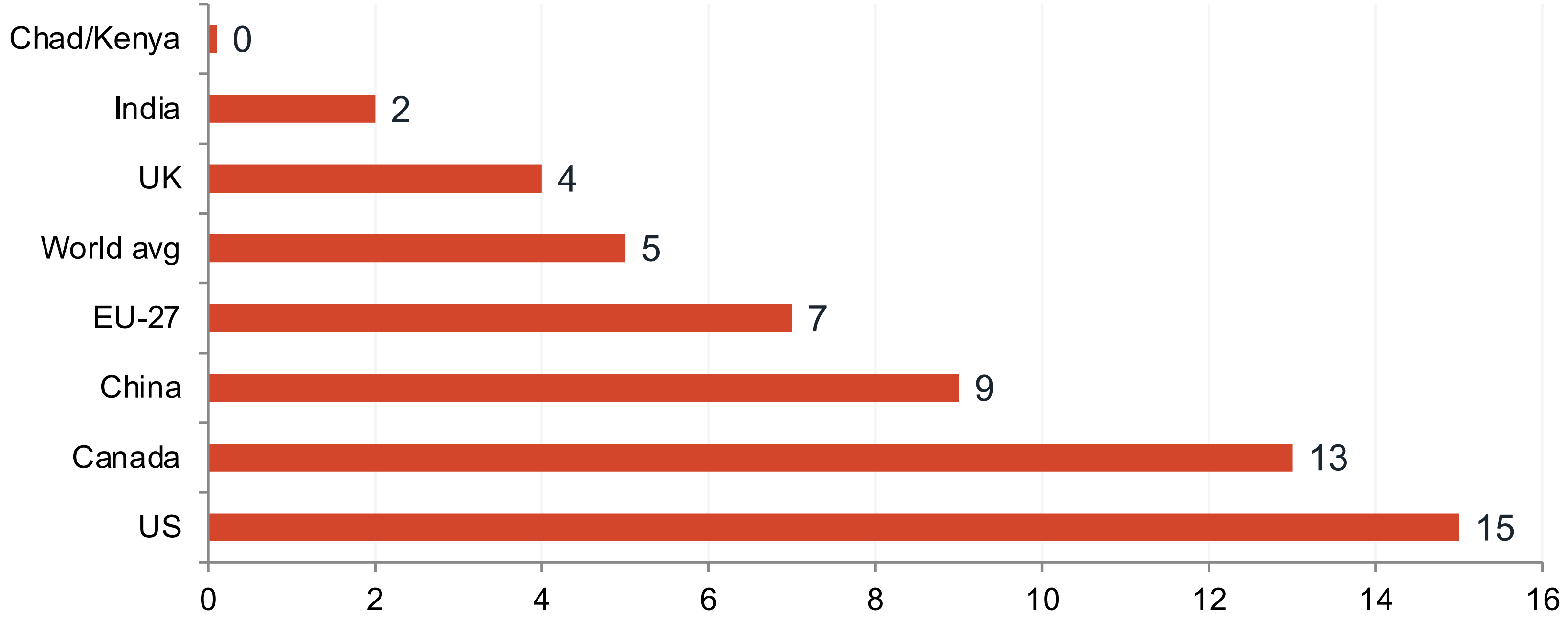
US + Europe combined have fallen below one-third of the global total



Africa and South America each contribute only 3–4% today, comparable to international aviation and shipping combined.

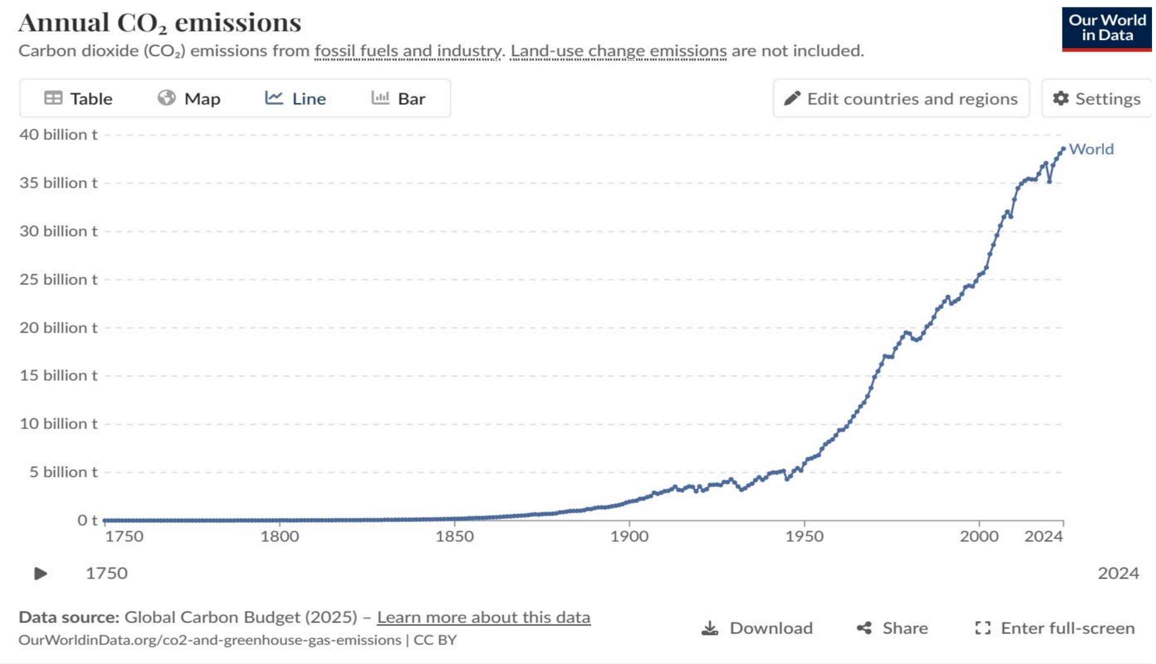
Per Capita Gap: A 150× Disparity

15 tonnes in the US vs. 0.1 tonnes in Chad



2024 Snapshot: Annual Change in Emissions

US and Europe Cut Emissions While Russia and SE Asia Increase



- Declines (Blue):
- • US and much of Europe: -1% to -5%

- Increases (Orange/Red):
- • Russia, parts of Africa, and SE Asia: +2% to +10%
- • Select South American nations: >10%

Energy Mix Drives Per Capita Differences

Similar Living Standards, Very Different Emissions Profiles

Country	Per Capita CO_2	Primary Energy Profile
France	~4 tonnes	High nuclear and renewables share
United Kingdom	~4 tonnes	High renewables, phasing out coal
Germany	~6-7 tonnes	Higher fossil fuel / coal share
Netherlands	~7 tonnes	Higher fossil fuel / natural gas reliance

Policy Insight:

Prosperity does not dictate a fixed level of emissions. A structural shift to nuclear and renewable energy can decouple high living standards from carbon intensity.

Key Takeaways

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

• Global emissions exceed 35 Bt and have not yet peaked, continuing an upward trajectory.

• China alone emits >25% of the global total, double that of the US, while India is rising rapidly.

• Per capita inequalities remain extreme, with a 150× disparity separating the highest emitters (US/Australia) from the lowest (Chad/Niger).

• US and Europe are declining in annual emissions, but retain massive historical cumulative responsibility.

• Policy and energy mix choices — not just prosperity — determine per capita outcomes, as seen in France vs. Germany